

PADHAISHURU

GATE Preparation Platform · padhaishuru.com

TREND-BASED MOCK PAPER

GATE Civil Engineering

Mock Paper — 03

Full-Length Examination Simulation

Duration: 3 Hours | Maximum Marks: 100 | Questions: 65

Examination Details

Branch:	Civil Engineering
Paper:	Mock 03
Total Questions:	65
Total Marks:	100
Duration:	3 Hours
Question Types:	MCQ, MSQ, NAT
Negative Marking:	Yes (MCQ only)
Difficulty:	Mixed

Unofficial practice material. Not affiliated with or endorsed by GATE/IIT.
Historical patterns do not guarantee future examination questions.
This is a trend-based practice simulation.

GENERAL INSTRUCTIONS

1. This question paper contains 65 questions for a total of 100 marks.
2. Duration of examination: 3 Hours.
3. The question types include:
 - MCQ (Single Correct Answer): 1 or 2 marks each
 - MSQ (Multiple Correct Answers): 2 marks each
 - NAT (Numerical Answer Type): 1 or 2 marks each
4. For MCQ: Only one option is correct. Select the single correct answer.
5. For MSQ: One or more options are correct. Select ALL correct options.
6. For NAT: Enter the numerical answer. No negative marking for NAT questions.
7. Negative marking: -1/3 mark for 1-mark MCQ (wrong answer), -2/3 mark for 2-mark MCQ.
No negative marking for MSQ and NAT questions.
8. A virtual calculator is available on the screen. Physical calculators are not permitted.
9. The question paper contains 8 pages including this instructions page.
10. All questions are compulsory.
11. Fill in the required details on the answer sheet before starting.
12. Do not ask for clarifications from the invigilator. Interpret questions as given.

Unofficial Practice Material — Not affiliated with or endorsed by GATE/IIT.

Prepared by PadhaiShuru (padhaishuru.com) for practice purposes only.

Section A: General Aptitude

Q1. [1-MCQ] [1 mark]

Volume of sphere r:

- (A) $\frac{4}{3} < \pi r^3$
- (B) $< \pi r^3$
- (C) $< \pi r^2$
- (D) $2 <$

Q2. [1-MCQ] [1 mark]

Choose similar to 'diligent':

- (A) Lazy
- (B) Industrious
- (C) Careless
- (D) Ignorant

Section B: Civil Engineering

Q3. [1-MCQ] [1 mark]

Order of $dy/dx=y$:

- (A) 0
- (B) 1
- (C) 2
- (D) 3

Q4. [1-MCQ] [1 mark]

For matrix $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$, $A^{50} * [1,0]^T$ equals:

- (A) $[1,0]^T$
- (B) $[2,1]^T$
- (C) $[(3+2\sqrt{2})^{50}, \dots]$

(D) Cannot determine

Q5. [2-MCQ] [2 marks]

A random variable $X \sim \text{Poisson}(\lambda=4)$. $P(X \geq 2)$ is approximately:

- (A) 0.908
- (B) 0.092
- (C) 0.762
- (D) 0.195

Q6. [2-MCQ] [2 marks]

For the ODE $dy/dx = y/x + \sqrt{y^2/x^2}$, with $y(1)=0$, the solution is:

- (A) $y = 0$
- (B) $y = x \ln(x)$
- (C) $y = x \sin(\ln x)$
- (D) $y = x^2$

Q7. [2-NAT] [2 marks]

$P(|Z| \leq 1)$ for $Z \sim N(0,1)$ is approximately:

(Numerical Answer Type — Enter the numerical value)

Q8. [2-MSQ] [2 marks]

For a Poisson distribution with mean 4, variance is:

- (A) 4
- (B) 2
- (C) 16
- (D) 1

Q9. [2-NAT] [2 marks]

Which of the following is a probability generating function for $\text{Poisson}(\lambda)$?

(Numerical Answer Type — Enter the numerical value)

Q10. [2-NAT] [2 marks]

$\lim_{n \rightarrow \infty} (1 + 1/n)^{2n}$ equals:

(Numerical Answer Type — Enter the numerical value)

Q11. [2-MCQ] [2 marks]

Fourier series of $f(x)=x$ on $(-\pi, \pi)$ has only:

- (A) Sine terms
- (B) Cosine terms
- (C) Both
- (D) Neither

Q12. [2-MCQ] [2 marks]

If $f(z) = z^2 + 3z + 2$ is analytic, then $\text{Re}(f(i)) =$

- (A) -1
- (B) -2
- (C) 1
- (D) 2

Q13. [2-MCQ] [2 marks]

Consider $f(x) = |x|$ on $[-1,1]$. This function is:

- (A) Differentiable everywhere
- (B) Continuous but not differentiable at 0
- (C) Neither continuous nor differentiable
- (D) Differentiable at 0 only

Q14. [2-NAT] [2 marks]

Laplace transform of $e^{(-3t)}u(t)$ is:

(Numerical Answer Type — Enter the numerical value)

Q15. [2-MCQ] [2 marks]

Rolle's theorem requires $f(x)$ to be:

- (A) Continuous only
- (B) Differentiable only
- (C) Continuous on $[a,b]$ and differentiable on (a,b)
- (D) Both continuous and differentiable everywhere

Q16. [2-MCQ] [2 marks]

The differential equation $y \, dx - x \, dy = 0$ has solution:

- (A) $y = C \cdot x$
- (B) $y = C/x$
- (C) $y = C \cdot x^2$
- (D) $x^2 + y^2 = C$

Q17. [2-MCQ] [2 marks]

If $f(x) = x^3 - 3x + 2$, number of real roots:

- (A) 1
- (B) 2
- (C) 3
- (D) 0

Q18. [2-NAT] [2 marks]

The area of a circle of radius r is:

(Numerical Answer Type — Enter the numerical value)

Q19. [2-NAT] [2 marks]

The Laplace transform of $t \cdot e^{(-2t)}$ is:

(Numerical Answer Type — Enter the numerical value)

Q20. [2-NAT] [2 marks]

The vector $(1,2,3)$ projected onto $(2,1,0)$ is:

(Numerical Answer Type — Enter the numerical value)

Q21. [2-NAT] [2 marks]

If a 3×3 matrix has eigenvalues 1, 2, and 3, then $\text{trace}(A^2)$ equals:

(Numerical Answer Type — Enter the numerical value)

Q22. [2-NAT] [2 marks]

$P(A \cup B) = 0.6$, $P(A \cap B) = 0.2$, $P(B) = 0.4$. Then $P(A|B)$ equals:

(Numerical Answer Type — Enter the numerical value)

Q23. [2-MCQ] [2 marks]

The rank of the matrix $\begin{bmatrix} 1, 2, 3 \\ 2, 4, 6 \\ 3, 6, 9 \end{bmatrix}$ is:

- (A) 1
- (B) 2
- (C) 3
- (D) 0

Q24. [2-NAT] [2 marks]

$\frac{d}{dx}[\ln(\sec x) + \ln(\tan x)]$ evaluated at $x = \pi/4$ equals:

(Numerical Answer Type — Enter the numerical value)

Q25. [2-NAT] [2 marks]

Gauss-Seidel method converges when the coefficient matrix is:

(Numerical Answer Type — Enter the numerical value)

Q26. [2-MCQ] [2 marks]

Cauchy-Schwarz inequality states $|\langle u, v \rangle| \leq$:

- (A) $\|u\| * \|v\|$
- (B) $\|u\| + \|v\|$
- (C) $\|u\| / \|v\|$
- (D) $\sqrt{\|u\| * \|v\|}$

Q27. [2-MCQ] [2 marks]

Number of spanning trees in complete graph K_4 :

- (A) 4
- (B) 8
- (C) 16
- (D) 12

Q28. [1-MCQ] [1 mark]

Galvanic cell converts:

- (A) Electrical to chemical
- (B) Chemical to electrical
- (C) Heat to work
- (D) Mechanical to electrical

Q29. [1-MCQ] [1 mark]

First-order half-life with $k=0.693 \text{ min}^{-1}$:

- (A) 1 min
- (B) 0.693 min
- (C) 1.44 min
- (D) 0.5 min

Q30. [2-MCQ] [2 marks]

pH of 10^{-8} M HCl approx:

- (A) 6
- (B) 7
- (C) 8
- (D) 6.98

Q31. [2-MCQ] [2 marks]

Strongest acid:

- (A) HCl
- (B) HNO₃
- (C) H₂SO₄
- (D) HClO₄

Q32. [2-MCQ] [2 marks]

$K_p = K_c(RT)^{\Delta n}$. For $N_2 + 3H_2 \rightleftharpoons 2NH_3$, $\Delta n =$

- (A) 2
- (B) -2
- (C) 1
- (D) -1

Q33. [1-NAT] [1 mark]

Max shear stress in solid shaft:

(Numerical Answer Type — Enter the numerical value)

Q34. [1-MCQ] [1 mark]

Cantilever UDL w over L . Deflection:

- (A) $wL^4/(8EI)$
- (B) $wL^4/(3EI)$
- (C) $5wL^4/(384EI)$
- (D) $PL^3/(3EI)$

Q35. [2-MSQ] [2 marks]

SS beam UDL w N/m, span L . Max BM:

- (A) $wL^2/8$
- (B) $wL^2/12$
- (C) $wL^2/4$
- (D) $wL^2/2$

Q36. [2-MCQ] [2 marks]

Fixed-free Euler buckling:

- (A) $\pi^2 EI/L^2$
- (B) $\pi^2 EI/(4L^2)$
- (C) $4\pi^2 EI/L^2$
- (D) $2\pi^2 EI/L^2$

Q37. [2-MCQ] [2 marks]

Cantilever UDL w , L . Fixed end moment:

- (A) $wL^2/2$
- (B) wL^2
- (C) $wL^2/4$
- (D) $wL^2/8$

Q38. [2-MCQ] [2 marks]

Influence line for SS beam reaction:

- (A) Triangle
- (B) Rectangle
- (C) Linear decreasing

(D) Parabola

Q39. [2-NAT] [2 marks]

Moment distribution DF for $K_{AB}=4EI/L$, $K_{AC}=3EI/L$:

(Numerical Answer Type — Enter the numerical value)

Q40. [1-MCQ] [1 mark]

Camber on road provides:

- (A) Speed
- (B) Drainage
- (C) Aesthetics
- (D) Strength

Q41. [2-NAT] [2 marks]

M25 concrete modulus of elasticity:

(Numerical Answer Type — Enter the numerical value)

Q42. [2-MCQ] [2 marks]

Slump test measures:

- (A) Strength
- (B) Workability
- (C) Durability
- (D) Setting time

Q43. [1-MCQ] [1 mark]

Gumbel distribution used for:

- (A) Rainfall
- (B) Flood frequency
- (C) Runoff
- (D) None

Q44. [1-MCQ] [1 mark]

Manning's equation for:

- (A) Pipe
- (B) Open channel
- (C) Laminar
- (D) Turbulent

Q45. [1-MCQ] [1 mark]

Darcy-Weisbach for:

- (A) Minor losses
- (B) Major losses
- (C) Both
- (D) None

Q46. [1-MCQ] [1 mark]

Rainfall measured by:

- (A) Hygrometer
- (B) Rain gauge
- (C) Anemometer
- (D) Barometer

Q47. [1-NAT] [1 mark]

LL=60%, PL=30%. PI =

(Numerical Answer Type — Enter the numerical value)

Q48. [1-MCQ] [1 mark]

Terzaghi q_{ult} for $c=0$, $\phi=30^\circ$, $B=2\text{m}$, $D_f=1\text{m}$, $\gamma=18$:

- (A) ~760 kPa
 - (B) ~450 kPa
 - (C) ~1800 kPa
 - (D) ~225 kPa
-

Q49. [2-NAT] [2 marks]

Rankine K_a for $\phi=30^\circ$:

(Numerical Answer Type — Enter the numerical value)

Q50. [1-MCQ] [1 mark]

Sum of first n odd numbers:

- (A) n^2
 - (B) $2n$
 - (C) $n(n+1)$
 - (D) $n(n-1)$
-

Q51. [1-MCQ] [1 mark]

Compound interest: P at $r\%$ for 2 years =

- (A) $P(1+r/100)^2$
 - (B) $P(1+2r/100)$
 - (C) $P + Pr/100 \times 2$
 - (D) $P \times r/100$
-

Q52. [2-MSQ] [2 marks]

If $x + 1/x = 3$, then $x^3 + 1/x^3 =$

- (A) 18
 - (B) 27
 - (C) 9
 - (D) 3
-

Q53. [1-MCQ] [1 mark]

Method of joints gives:

- (A) Forces only
 - (B) Moments
 - (C) Both
 - (D) None
-

Q54. [1-MCQ] [1 mark]

Bending moment in UDL simply supported:

- (A) $wL^2/8$
- (B) $wL^2/12$
- (C) $wL/2$
- (D) wL^2

Q55. [1-MCQ] [1 mark]

Moment-area method gives:

- (A) Slope
- (B) Deflection
- (C) Both
- (D) None

Q56. [1-NAT] [1 mark]

Bearing capacity of footing:

(Numerical Answer Type — Enter the numerical value)

Q57. [1-MCQ] [1 mark]

Atterberg limits:

- (A) Liquid limit
- (B) Plastic limit
- (C) Shrinkage
- (D) All

Q58. [1-NAT] [1 mark]

BOD stands for:

(Numerical Answer Type — Enter the numerical value)

Q59. [1-MCQ] [1 mark]

Sedimentation removes:

- (A) Dissolved
- (B) Suspended
- (C) Colloids
- (D) Gases

Q60. [1-MCQ] [1 mark]

PCU stands for:

- (A) A
- (B) B
- (C) C
- (D) D

Q61. [1-MCQ] [1 mark]

Camber provided for:

- (A) Strength
- (B) Drainage
- (C) Smoothness
- (D) Aesthetics

Q62. [1-MCQ] [1 mark]

Theodolite measures:

- (A) Distance
- (B) Angle
- (C) Level
- (D) All

Q63. [1-NAT] [1 mark]

Dumpy level used for:

(Numerical Answer Type — Enter the numerical value)

Q64. [1-NAT] [1 mark]

Initial setting time OPC:

(Numerical Answer Type — Enter the numerical value)

Q65. [1-MCQ] [1 mark]

Water-cement ratio affects:

- (A) Strength only
 - (B) Workability only
 - (C) Both
 - (D) None
-

ANSWER KEY

Q. No.	Type	Answer	Marks
1	1MCQ	A	1
2	1MCQ	B	1
3	1MCQ	B	1
4	1MCQ	A	1
5	1MCQ	A	1
6	1MCQ	C	1
7	1NAT	C	1
8	1MCQ	D	1
9	1MCQ	B	1
10	1MCQ	B	1
11	1MCQ	B	1
12	1MCQ	B	1
13	1NAT	A	1

14	1MCQ	B	1
15	1MCQ	A	1
16	1MCQ	B	1
17	1MCQ	B	1
18	1NAT	C	1
19	1NAT	A	1
20	1MCQ	C	1
21	1MCQ	B	1
22	1NAT	A	1
23	1MCQ	A	1
24	1MCQ	A	1
25	1MCQ	A	1
26	1MCQ	B	1
27	1NAT	A	1
28	1MCQ	B	1

29	1MCQ	A	1
30	1MCQ	A	1
31	2MCQ	A	2
32	2MSQ	A	2
33	2MCQ	A	2
34	2NAT	A	2
35	2MCQ	A	2
36	2MSQ	A	2
37	2MCQ	C	2
38	2NAT	A	2
39	2MCQ	D	2
40	2NAT	A	2
41	2MCQ	B	2
42	2MSQ	A	2
43	2MCQ	B	2

44	2NAT	A	2
45	2NAT	B	2
46	2MCQ	A	2
47	2NAT	A	2
48	2MCQ	A	2
49	2MCQ	D	2
50	2MCQ	B	2
51	2MCQ	B	2
52	2NAT	A	2
53	2MCQ	C	2
54	2MCQ	B	2
55	2MCQ	B	2
56	2NAT	A	2
57	2NAT	A	2
58	2NAT	B	2

59	2NAT	A	2
60	2NAT	B	2
61	2MCQ	A	2
62	2NAT	C	2
63	2NAT	C	2
64	2MCQ	A	2
65	2MCQ	C	2

DETAILED SOLUTIONS

Q1.

Answer: A

Solution:

Correct Answer: A

Subject: General Aptitude | Topic: Quantitative | Difficulty: difficult

Q2.

Answer: B

Solution:

Correct Answer: B

Subject: General Aptitude | Topic: Verbal | Difficulty: moderate

Q3.

Answer: B

Solution:

Correct Answer: B

Subject: Engineering Mathematics | Topic: Differential Equations | Difficulty: easy

Q4.

Answer: A

Solution:

Correct Answer: A

Subject: Structural Engineering | Topic: Trusses | Difficulty: difficult

Q5.

Answer: A

Solution:

Correct Answer: A

Subject: Structural Engineering | Topic: Beams | Difficulty: easy

Q6.

Answer: C

Solution:

Correct Answer: C

Subject: Structural Engineering | Topic: Deflection | Difficulty: moderate

Q7.

Answer: C

Solution:

Correct Answer: C

Subject: Geotechnical Engineering | Topic: Foundation | Difficulty: moderate

Q8.

Answer: D

Solution:

Correct Answer: D

Subject: Geotechnical Engineering | Topic: Soil Mechanics | Difficulty: easy

Q9.

Answer: B

Solution:

Correct Answer: B

Subject: Water Resources Engineering | Topic: Flood | Difficulty: difficult

Q10.

Answer: B

Solution:

Correct Answer: B

Subject: Water Resources Engineering | Topic: Open Channel | Difficulty: moderate

Q11.

Answer: B

Solution:

Correct Answer: B

Subject: Water Resources Engineering | Topic: Pipe Flow | Difficulty: easy

Q12.

Answer: B

Solution:

Correct Answer: B

Subject: Water Resources Engineering | Topic: Precipitation | Difficulty: easy

Q13.

Answer: A

Solution:

Correct Answer: A

Subject: Environmental Engineering | Topic: Sewage | Difficulty: moderate

Q14.

Answer: B

Solution:

Correct Answer: B

Subject: Environmental Engineering | Topic: Water | Difficulty: moderate

Q15.

Answer: A

Solution:

Correct Answer: A

Subject: Transportation Engineering | Topic: Traffic | Difficulty: easy

Q16.

Answer: B

Solution:

Correct Answer: B

Subject: Transportation Engineering | Topic: Highways | Difficulty: moderate

Q17.

Answer: B

Solution:

Correct Answer: B

Subject: Surveying & Geomatics | Topic: Theodolite | Difficulty: easy

Q18.

Answer: C

Solution:

Correct Answer: C

Subject: Surveying & Geomatics | Topic: Levelling | Difficulty: easy

Q19.

Answer: A

Solution:

Correct Answer: A

Subject: Construction Materials | Topic: Cement | Difficulty: easy

Q20.

Answer: C

Solution:

Correct Answer: C

Subject: Construction Materials | Topic: Mix Design | Difficulty: moderate

Q21.

Answer: B

Solution:

Galvanic: spontaneous redox converts chemical to electrical energy.

Subject: Chemistry | Topic: Chemistry | Difficulty: moderate

Q22.

Answer: A

Solution:

$$PI = LL - PL = 60 - 30 = 30.$$

Subject: Geotechnical | Topic: Geotechnical | Difficulty: moderate

Q23.

Answer: A

Solution:

$$q_{ult} = q \cdot N_q + 0.5 \cdot \gamma \cdot B \cdot N_\gamma = 18 \cdot 22.5 + 0.5 \cdot 18 \cdot 2 \cdot 19.7 = 405 + 354.6 = 759.6 \text{ kPa}.$$

Subject: Geotechnical | Topic: Geotechnical | Difficulty: moderate

Q24.

Answer: A

Solution:

$$1+3+5+\dots+(2n-1) = n^2.$$

Subject: Miscellaneous | Topic: Miscellaneous | Difficulty: difficult

Q25.

Answer: A

Solution:

$$CI = P(1+r/100)^n. \text{ For 2 years: } P(1+r/100)^2.$$

Subject: Miscellaneous | Topic: Miscellaneous | Difficulty: moderate

Q26.

Answer: B

Solution:

Camber: cross slope for rainwater drainage.

Subject: Civil Engineering | Topic: Civil Engineering | Difficulty: moderate

Q27.

Answer: A

Solution:

$$\tau_{max} = T \cdot c/J = 16T/(\pi \cdot d^3).$$

Subject: SOM | Topic: SOM | Difficulty: easy

Q28.

Answer: B

Solution:

$$\text{Cantilever UDL: } \delta = wL^4/(8EI).$$

Subject: SOM | Topic: SOM | Difficulty: moderate

Q29.

Answer: A

Solution:

Eigenvalues are 3 and 1. $[1,0]^T$ is eigenvector for eigenvalue 1. $A^{50} \cdot [1,0]^T = [1,0]^T$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q30.

Answer: A

Solution:

$$t_{1/2} = 0.693/k = 0.693/0.693 = 1 \text{ min}.$$

Subject: Chemistry | Topic: Chemistry | Difficulty: moderate

Q31.

Answer: A

Solution:

$$P(X \geq 2) = 1 - e^{-4(1+4)} = 1 - 5e^{-4} \approx 0.908.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q32.

Answer: A

Solution:

$$\text{Cube: } (x+1/x)^3 = x^3 + 3x + 3/x + 1/x^3 = 27. \quad x^3 + 1/x^3 = 27 - 3 \cdot 3 = 18.$$

Subject: Miscellaneous | Topic: Miscellaneous | Difficulty: easy

Q33.

Answer: A

Solution:

FEM = $wL \cdot (L/2) = wL^2/2$ (hogging).

Subject: Structural Analysis | Topic: Structural Analysis | Difficulty: moderate

Q34.

Answer: A

Solution:

$K_a = \tan^2(45-15) = \tan^2(30) = 1/3$.

Subject: Geotechnical | Topic: Geotechnical | Difficulty: moderate

Q35.

Answer: A

Solution:

Homogeneous ODE. Substituting $y=x \cdot v$ and $y(1)=0$ gives $y=0$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q36.

Answer: A

Solution:

SS UDL: max BM at center = $wL^2/8$.

Subject: SOM | Topic: SOM | Difficulty: moderate

Q37.

Answer: C

Solution:

$R_A = (L-x)/L$. Linear decreasing from L to 0.

Subject: Structural Analysis | Topic: Structural Analysis | Difficulty: easy

Q38.

Answer: A

Solution:

Standard normal: $P(-1 < Z < 1)$ approx 0.68.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q39.

Answer: D

Solution:

Very dilute: $[H^+]$ from water dominates. $[H^+]_{total}$ approx $1.1e-7$. pH approx 6.96.

Subject: Chemistry | Topic: Chemistry | Difficulty: difficult

Q40.

Answer: A

Solution:

$E_c = 5000 \cdot \sqrt{f_{ck}} = 5000 \cdot 5 = 25000$ MPa.

Subject: Civil Engineering | Topic: Civil Engineering | Difficulty: difficult

Q41.

Answer: B

Solution:

$K=2$, $L_{eff}=2L$. $P_{cr} = \pi^2 EI / (2L)^2 = \pi^2 EI / (4L^2)$.

Subject: SOM | Topic: SOM | Difficulty: easy

Q42.

Answer: A

Solution:

Poisson: mean = lambda = variance. Both = 4.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q43.

Answer: B

Solution:

Slump: workability/consistency of fresh concrete.

Subject: Civil Engineering | Topic: Civil Engineering | Difficulty: easy

Q44.

Answer: A

Solution:

PGF of Poisson(λ): $G(z) = E[z^X] = \exp(\lambda(z-1))$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q45.

Answer: B

Solution:

$(1+1/n)^{2n} = [(1+1/n)^n]^2 \rightarrow e^2$ as $n \rightarrow \infty$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q46.

Answer: A

Solution:

$f(x)=x$ is odd on symmetric interval. Fourier series has only sine terms.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q47.

Answer: A

Solution:

$DF_{AB} = 4/(4+3) = 4/7 = 0.57$.

Subject: Structural Analysis | Topic: Structural Analysis | Difficulty: moderate

Q48.

Answer: A

Solution:

$f(i) = i^2 + 3i + 2 = -1 + 3i$. $\text{Re}(f(i)) = -1$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q49.

Answer: D

Solution:

Perchloric acid (HClO_4) is the strongest among common acids.

Subject: Chemistry | Topic: Chemistry | Difficulty: moderate

Q50.

Answer: B

Solution:

$|x|$ is continuous everywhere. Left derivative at 0 = -1, right = +1. Not differentiable at 0.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q51.

Answer: B

Solution:

$\Delta_n = 2 - 4 = -2$.

Subject: Chemistry | Topic: Chemistry | Difficulty: difficult

Q52.

Answer: A

Solution:

$L\{e^{-at}\} = 1/(s+a)$. Here $a=3$: $1/(s+3)$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q53.

Answer: C

Solution:

Rolle's: f continuous on $[a,b]$, differentiable on (a,b) , $f(a)=f(b)$. Then $f'(c)=0$ for some c in (a,b) .

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q54.

Answer: B

Solution:

$dy/y = dx/x$. $\ln(y) = \ln(x) + C$. $y = C \cdot x$. Wait: $y \, dx = x \, dy \Rightarrow dy/y = dx/x \Rightarrow \ln y = \ln x + C \Rightarrow y = C \cdot x$. That's option A. Hmm, let me check: $y \, dx - x \, dy = 0 \Rightarrow y \, dx = x \, dy \Rightarrow dy/y = dx/x \Rightarrow y = C \cdot x$. Answer: A.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q55.

Answer: B

Solution:

$f(0)=2$, $f(1)=0$, $f(-1)=4$, $f(2)=4$. $f(-2)=-8+6+2=0$. Roots at $x=1$ (double?) and $x=-2$. $f(x)=(x-1)^2(x+2)$. Two distinct real roots.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q56.

Answer: A

Solution:

Area = πr^2 . Circumference = $2\pi r$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q57.

Answer: A

Solution:

$L\{f(t)\} = -d/ds(F(s))$. $F(s) = 1/(s+2)$. $-d/ds[1/(s+2)] = 1/(s+2)^2$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q58.

Answer: B

Solution:

$\text{proj} = ((v \cdot w)/(w \cdot w)) * w = (4/5) * (2, 1, 0) = (8/5, 4/5, 0)$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q59.

Answer: A

Solution:

Eigenvalues of A^2 are 1, 4, 9. Trace = sum = $1+4+9 = 14$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q60.

Answer: B

Solution:

$P(A|B) = P(A \cap B)/P(B) = 0.2/0.4 = 0.5$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q61.

Answer: A

Solution:

Row 2 = $2 \times$ Row 1, Row 3 = $3 \times$ Row 1. All rows linearly dependent. Rank = 1.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q62.

Answer: C

Solution:

$d/dx[\ln(\sin x / \cos^2 x)] = \cot x + 2 \tan x$. At $\pi/4$: $1 + 2 = 3$. Hmm. Let me recalc: $d/dx \ln(\sec x) = \tan x$. $d/dx \ln(\tan x) = \sec^2 x / \tan x = 1/(\sin x \cos x) = 2 \csc(2x)$. At $\pi/4$: $1 + 2 = 3$. Not in options. The derivative of $\ln(\sec x * \tan x) = d/dx[\ln(\sin x / \cos^2 x)] = \cot x + 2 \tan x$. At $\pi/4$: $1+2=3$. Hmm.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q63.

Answer: C

Solution:

Gauss-Seidel converges if matrix is diagonally dominant. Jacobi needs symmetric positive definite.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q64.

Answer: A

Solution:

$|\langle u, v \rangle| \leq \|u\| * \|v\|$. Equality when u and v are linearly dependent.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q65.

Answer: C

Solution:

Kirchhoff: for K_n , spanning trees = $n^{n-2} = 4^2 = 16$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

PAPER BLUEPRINT

Subject Distribution:

Engineering Mathematics: 25 questions, 48 marks
Chemistry: 5 questions, 8 marks
SOM: 4 questions, 6 marks
Structural Analysis: 3 questions, 6 marks
Civil Engineering: 3 questions, 5 marks
Water Resources Engineering: 4 questions, 4 marks
Geotechnical: 3 questions, 4 marks
Miscellaneous: 3 questions, 4 marks
Structural Engineering: 3 questions, 3 marks
General Aptitude: 2 questions, 2 marks
Geotechnical Engineering: 2 questions, 2 marks
Environmental Engineering: 2 questions, 2 marks
Transportation Engineering: 2 questions, 2 marks
Surveying & Geomatics: 2 questions, 2 marks
Construction Materials: 2 questions, 2 marks

Difficulty Distribution:

Easy: 20 | Moderate: 35 | Difficult: 10

Statistics:

Total Questions: 65
Total Marks: 100

This blueprint is generated from the actual question distribution in this paper.